

An Approximate Capacity Bound for Sparse Combinatorial Key-Value Memory: Full Derivation and Validation

Supplementary note to “Combinatorial Coding, Key-Value Recall, and Emergent
Constraint Satisfaction in a Hippocampal-Inspired Spiking Network”

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Abstract

This note gives the complete derivation, with every intermediate step made explicit, of the approximate capacity bound used in the main paper (Section 2.3): an expression for the largest number of items M a sparse K -of- H combinatorial key-value memory can store while remaining reliably retrievable under a given level of query corruption. The bound is structurally the same as modern Hopfield networks’ proven exponential capacity [1], but is derived here for a discrete, sparse code via elementary combinatorics and extreme value theory rather than for dense continuous patterns via an energy-function argument. We include a full, self-contained proof of the one non-elementary step (the expected maximum of many correlated-in-effect-but-treated-as-independent Gaussian similarities), validate the final bound against simulation, and discuss its relationship to the wider family of capacity theorems it belongs to.

1 Setup

We store M items. Item m ’s key is a binary vector $h_m \in \{0, 1\}^H$ with exactly K active (value-1) coordinates – a K -of- H sparse code. Throughout this note we model the M stored keys as independent, uniformly random K -subsets of $[H] = \{1, \dots, H\}$: an idealization relative to the true architecture, where keys are W_{SH} -projections of mutually correlated combinatorial structure (solved Sudoku grids), addressed explicitly in Section 8.

A query is formed from the true key h_{true} by flipping a fraction $\eta \in [0, 1]$ of all H coordinates, chosen uniformly at random without replacement – the noise model used throughout the main paper. Retrieval compares the query against all M stored keys via similarity (shared active coordinates) and should return the index of h_{true} . Retrieval *fails* when some other stored key’s similarity to the query exceeds h_{true} ’s.

We ask: for fixed H , K , and corruption level η , how large can M be before this kind of failure becomes likely?

2 Step 1: Similarity between two independent random keys

Lemma 1 (Hypergeometric overlap). *Let $A, B \subset [H]$ be independent uniformly random subsets with $|A| = a$, $|B| = b$. Then $|A \cap B|$ is a hypergeometric random variable with*

$$\mathbb{E}[|A \cap B|] = \frac{ab}{H}, \quad \text{Var}(|A \cap B|) = a \cdot \frac{b}{H} \left(1 - \frac{b}{H}\right) \cdot \frac{H - a}{H - 1}.$$

Proof sketch. Fix A and treat B 's b elements as drawn sequentially without replacement from $[H]$. Let $X_i = \mathbb{I}[\text{the } i\text{-th draw lands in } A]$, so $|A \cap B| = \sum_{i=1}^b X_i$. Each X_i is Bernoulli(a/H) by symmetry, giving the mean by linearity of expectation. For the variance, $\text{Cov}(X_i, X_j) < 0$ for $i \neq j$ because sampling without replacement makes draws negatively associated (having drawn a marked element makes drawing another marked element on a later draw slightly less likely); summing the b variances and $\binom{b}{2}$ covariances gives the stated finite-population-corrected formula. (Full derivation: Feller, *An Introduction to Probability Theory*, Vol. 1, Ch. II.) $\square$

Applying Lemma 1 with $a = b = K$ gives the similarity between two *uncorrupted* stored keys: mean K^2/H , the number quoted throughout the main paper (e.g. $K^2/H = 22.5$ for the default $H = 1000$, $K = 150$).

3 Step 2: Statistics of a corrupted query

Flipping a uniformly random fraction η of all H coordinates is itself an instance of Lemma 1: the ηH flipped positions are a random subset of $[H]$, and we ask how many of h_{true} 's K active positions they contain. By the lemma (with $a = K$, $b = \eta H$), the expected number of active positions flipped *off* is ηK , so

$$S_{\text{true}}(\eta) = \mathbb{E}[\text{similarity of query to } h_{\text{true}}] = K(1 - \eta), \quad (1)$$

exactly (no further approximation needed: this is a direct expectation, not a bound). The expected number of originally-*inactive* positions flipped *on* is, symmetrically, $\eta(H - K)$, so the corrupted query's own total active-coordinate count is, in expectation,

$$K_q(\eta) = K(1 - \eta) + \eta(H - K) = K(1 - 2\eta) + \eta H. \quad (2)$$

The query's similarity to any *other*, independent stored key is now Lemma 1 applied with $a = K_q(\eta)$, $b = K$:

$$\mu_{\text{other}}(\eta) = \frac{K_q(\eta) K}{H}, \quad \sigma_{\text{other}}^2(\eta) = K_q(\eta) \frac{K}{H} \left(1 - \frac{K}{H}\right) \frac{H - K_q(\eta)}{H - 1}. \quad (3)$$

4 Step 3: Gaussian approximation

The hypergeometric distribution of S_{other} (similarity to one specific other key) is exact but discrete and unwieldy for the extreme-value step that follows. We approximate

$$S_{\text{other}} \approx \mathcal{N}(\mu_{\text{other}}(\eta), \sigma_{\text{other}}^2(\eta)),$$

justified for $H \gg 1$ by the central limit theorem for sampling without replacement from a finite population [2, 3]: sums arising from simple random sampling without replacement are asymptotically normal as the population size grows, with exactly the finite-population-corrected variance already appearing in Lemma 1. This is the least exact step in the derivation, and its error is largest precisely when σ_{other} is small relative to the mean – i.e. at low noise, where the discreteness of the underlying hypergeometric distribution matters most. Section 7 quantifies this error directly rather than leaving it as a caveat.

5 Step 4: The maximum of many competitors

Retrieval does not fail when the *typical* competing key looks similar to the query – it fails when the *single most similar* of the $M - 1$ competitors exceeds $S_{\text{true}}(\eta)$. This is an extreme-value criterion, and it is the step that most distinguishes this derivation from a naive signal-to-noise argument.

Lemma 2 (Expected maximum of i.i.d. Gaussians). *Let $Z_1, \dots, Z_n$ be i.i.d. $\mathcal{N}(0, 1)$. Then*

$$\mathbb{E} \left[\max_{1 \leq i \leq n} Z_i \right] \leq \sqrt{2 \ln n} + o(1) \quad \text{as } n \rightarrow \infty,$$

and this bound is asymptotically tight: $\mathbb{E}[\max_i Z_i] = \sqrt{2 \ln n} (1 + o(1))$.

Proof of the upper bound. For any $t \geq 0$, the standard Gaussian tail (Chernoff) bound gives $\Pr(Z_1 > t) \leq e^{-t^2/2}$: with $s = t$, $\Pr(Z_1 > t) = \Pr(e^{sZ_1} > e^{st}) \leq \mathbb{E}[e^{sZ_1}]e^{-st} = e^{s^2/2-st}$, minimized at $s = t$. A union bound over the n variables gives $\Pr(\max_i Z_i > t) \leq \min(1, ne^{-t^2/2})$. Writing the expectation of a nonnegative random variable as an integral of its tail,

$$\mathbb{E} \left[\max_i Z_i \right] = \int_0^\infty \Pr \left(\max_i Z_i > t \right) dt \leq \int_0^{t^*} 1 dt + \int_{t^*}^\infty ne^{-t^2/2} dt,$$

where $t^* = \sqrt{2 \ln n}$ is chosen so the two bounds on the integrand meet ($ne^{-(t^*)^2/2} = 1$). The first integral is exactly $t^* = \sqrt{2 \ln n}$. For the second, the standard Gaussian tail estimate $\int_a^\infty e^{-t^2/2} dt \leq \frac{1}{a} e^{-a^2/2}$ (for $a > 0$, obtained by writing $e^{-t^2/2} \leq \frac{t}{a} e^{-t^2/2}$ on $t \geq a$ and integrating exactly) gives

$$\int_{t^*}^\infty ne^{-t^2/2} dt \leq \frac{n}{t^*} e^{-(t^*)^2/2} = \frac{1}{\sqrt{2 \ln n}} \xrightarrow{n \rightarrow \infty} 0.$$

Summing the two pieces gives $\mathbb{E}[\max_i Z_i] \leq \sqrt{2 \ln n} + o(1)$. The matching lower bound requires a second-moment (Paley–Zygmund-type) argument and is omitted here; see Boucheron, Lugosi & Massart, *Concentration Inequalities* [4], Ch. 2, for the complete two-sided result. $\square$

Rescaling Lemma 2 to $\mathcal{N}(\mu_{\text{other}}, \sigma_{\text{other}}^2)$ variables and setting $n = M - 1$ (the number of competing stored keys) gives the working approximation used in the main paper:

$$\mathbb{E} \left[\max_{i=1}^{M-1} S_{\text{other},i} \right] \approx \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}. \quad (4)$$

6 Step 5: Solving for capacity

Retrieval is (approximately, at the level of expectations rather than a proper tail bound on failure probability) reliable when $S_{\text{true}}(\eta)$ exceeds the expected maximum competitor, Equation 4. Define the normalized separation margin

$$z(\eta) = \frac{S_{\text{true}}(\eta) - \mu_{\text{other}}(\eta)}{\sigma_{\text{other}}(\eta)}. \quad (5)$$

Setting $S_{\text{true}}(\eta) = \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}$ and solving for M :

$$z(\eta) = \sqrt{2 \ln(M-1)} \iff M-1 = \exp\left(\frac{z(\eta)^2}{2}\right) \iff \boxed{M_{\text{max}}(\eta) \approx 1 + \exp\left(\frac{z(\eta)^2}{2}\right)}. \quad (6)$$

Proposition 1 (Structural form). *Equation 6 states that capacity is exponential in the square of a normalized signal-to-noise-like quantity. This is the same structural form as modern Hopfield networks’ proven capacity $C \cong 2^{d/2}$ [1], though that bound is derived via a Lyapunov energy-function argument (the Concave–Convex Procedure applied to a log-sum-exp energy) over dense continuous patterns, not via hypergeometric statistics over a sparse discrete code.*

7 Validation against simulation

Equations 1–6 were implemented exactly as derived (`capacity_bound_derivation.py`, no free parameters fit to data) and compared against the empirically measured collapse thresholds reported in the main paper.

	$M=50$	$M=1000$	$M=10,000$	$K=10$	$K=150$
Predicted (this note, $H=1000$)	43.9%	41.9%	40.6%	24.5%	41.9%
Observed (simulation)	45%	45%	40%	25%	40–45%

Agreement is within 1–5 percentage points across a 200-fold range of M and an order of magnitude in K , including an almost exact match at $K = 10$. A second, independent check: at $H=1000$, $K=150$, $M=200$, the derivation predicts the ratio $S_{\text{true}}(\eta)/\mathbb{E}[\max_i S_{\text{other},i}]$ at $\eta \in \{0, 0.20, 0.30\}$ as 4.21, 1.99, 1.47 respectively, against independently measured ratios of 3.39, 1.92, 1.45 (obtained from a completely separate experiment, the separation-operator ablation, not used to fit any parameter of this derivation). The two noisy points match to within 4% and 1.4%; the noise-free point is over-predicted by about 24%, consistent with Section 4’s observation that the Gaussian approximation is weakest exactly there.

8 Limitations

Two assumptions are false by construction and are stated as such rather than absorbed silently.

Independence. Real stored keys are W_{SH} -projections of genuinely correlated combinatorial objects (solved Sudoku grids share row/column/box structure), not independent uniform random subsets. The observed collapse thresholds sit consistently at or slightly above the predicted ones (Section 7); a plausible explanation is that this real correlation makes the worst-case competing key marginally less extreme than the independent model assumes, though this has not been tested directly.

The extreme-value step is a first-moment approximation, not a tail bound. Equation 4 matches the *expected* maximum, not a high-probability bound on it; a fully rigorous capacity statement would instead bound $\Pr(\max_i S_{\text{other},i} > S_{\text{true}}(\eta))$ directly (e.g. via the same union-bound technique used in the proof of Lemma 2, without first taking an expectation) and would very likely give a somewhat smaller M_{max} at any fixed, small target failure probability. The bound derived here should be read as characterizing the right *mechanism and order of magnitude* – exponential-in-margin capacity, hence a sharp rather than gradual collapse, hence a threshold only weakly dependent on M – not as a proof.

9 Relation to the wider literature

This derivation belongs to a family of results sharing one underlying question: does a correct answer stand out against the *extreme* value among exponentially or combinatorially many near-random

alternatives? The same skeleton, worked out through different technical routes, underlies Shannon’s channel coding theorem [5], the Random Energy Model in spin-glass theory [6], Gardner’s perceptron capacity [7], and the classical Amit–Gutfreund–Sompolinsky analysis of Hopfield network capacity [8]. Kanerva’s original capacity analysis for sparse distributed memory [9] is the closest existing derivation in spirit to this one – also a sparse binary code, also analyzed via combinatorial (there, Hamming-ball volume) arguments – though via a different geometric route than the hypergeometric-overlap statistics used here.

References

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